

ABU HASNAT ABDULLAH

Senior Machine Learning Engineer | LLM & Agentic Systems, MLOps, Time-Series & Computer Vision

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SUMMARY

Machine learning engineer with 6+ years building production ML and LLM systems in Python for clients in Norway, Sweden, and Bangladesh. Works across the full lifecycle: data pipelines and feature stores, model training and fine-tuning, evaluation and guardrails, deployment, and monitoring on Azure and AWS. Now lead developer on Sensa AS's industrial-AI platform: RAG and agentic LLM systems, time-series anomaly detection, and computer vision.

- Raised object-detection accuracy from 40% to 98% with a YOLOv11 model on real-time aquaculture video.
- Fine-tuned a local 3B LLM (QLoRA) to 97% accuracy, matching cloud-API quality at far lower cost.
- Architected a RAG + agent assistant over a knowledge graph and live time-series data for operators.
- Led teams of up to 8 engineers across 6 client projects; shipped production services used by 500+ users.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, YOLOv11, OpenCV

LLM & Agents: RAG, LangChain, LangGraph, MCP, OpenAI, QLoRA fine-tuning, LLM evals, guardrails

Applied ML: Time-series forecasting, anomaly detection, computer vision, OCR, NLP, feature engineering

MLOps: MLflow, Azure ML, model registry, drift monitoring, Docker, CI/CD (GitHub Actions), Mage.ai pipelines

Data: Python, SQL, Pandas, NumPy, PostgreSQL, InfluxDB, Neo4j, PGVector, Chroma, MongoDB

Cloud & Serving: Azure, AWS (Lambda, ECS, EC2, S3), FastAPI, REST APIs, Kafka, Redis

Practices: AI-assisted development (Claude Code), TDD, code review, mentoring, Agile

EXPERIENCE

Senior Software Engineer — Cefalo Bangladesh Ltd.

Nov 2024 – Present

Lead developer for Sensa AS (Norway) and internal AI products · AI Task Force · AI Hackathon mentor & judge

- **Unity AI:** Architected Unity AI, an LLM assistant that lets industrial plant operators query live plant data in plain language: hybrid RAG over a Neo4j knowledge graph, InfluxDB time-series data, and PGVector, with LangGraph agents, MCP tools, and source-cited answers under data-governance rules to limit hallucinations.
- **Unity MLOps:** Built an end-to-end anomaly-detection platform for time-series sensor data: automated training with Isolation Forest and the pretrained Moirai v2 model, MLflow experiment tracking and model registry, Azure ML deployment, and drift monitoring, turning manual log review into real-time alerts.
- **Pellet Detection:** Trained and deployed a YOLOv11-nano object-detection model that detects and counts fish-feed pellets in real-time aquaculture video, raising detection accuracy from 40% to 98% and enabling automated feed-rate optimization for sensor-equipped fish farms (PyTorch, OpenCV, Docker).
- **CVInsight:** Led an internal LLM résumé parser that turns PDF and DOCX CVs into structured, ATS-ready data with role and skill categories, plus a recruiter Q&A chatbot. Fine-tuned and benchmarked Llama 3.2, DeepSeek, and Qwen with QLoRA (Unsloth); the locally hosted Llama 3.2 3B reached 97% accuracy, matching cloud-API quality at a fraction of the cost.
- **Unity Engine:** Built the data-orchestration and feature platform behind Unity AI, Unity MLOps, and Pellet Detection: Mage.ai ETL pipelines ingest high-throughput sensor and external data into time-series and feature stores, with reusable transforms shared by real-time anomaly detection and RAG retrieval.
- **AI-assisted delivery:** Shipped production services used by 500+ users through an AI-agent-led development workflow (Claude Code) with agent harnesses, guardrails, automated testing, and end-to-end monitoring.

Tech: Python, PyTorch, YOLOv11, OpenCV, scikit-learn, LangGraph, MCP, Unsloth, MLflow, Azure ML, Neo4j, InfluxDB

Senior Software Engineer — Strativ AB

Apr 2021 – Jan 2024

Tech lead for client products in Sweden: Resemolnets, Disruptive Ventures, Kry

- **Team leadership:** Led cross-functional teams of up to 8 engineers across 6 client projects, owning system design, backend architecture, code review, and delivery while staying hands-on in API and AI features.
- **Julia AI (TourGPT):** Architected a SaaS travel chatbot for Resemolnets, grounded in a custom tour knowledge base, that gives personalized tour recommendations and handles orders end to end. Owned prompt engineering, embedding strategy (PGVector), and safety guardrails from prototype to production.
- **Investment platform:** Designed a platform for Disruptive Ventures that tracks portfolio-company performance, sales, and accounting; integrated third-party accounting and marketing data through event-driven processing (Kafka, Redis) and added time-series sales forecasting with scikit-learn.
- **Health Check:** Built Kry's digital primary-care platform ingesting body measurements and lab blood-test results from third-party APIs, with predictive health-condition analysis in scikit-learn.

Tech: Python, scikit-learn, Pandas, OpenAI, LangChain, PGVector, Django, PostgreSQL, Kafka, Redis, AWS, React

Software Engineer — Circle Fintech Ltd.

Sep 2020 – Apr 2021

- **Boarding Bay:** Built 3 production microservices (national ID OCR, face detection, and billing) for Jamuna Bank's eKYC platform for customer onboarding and business transactions, automating identity checks that were previously done by hand.
- Applied OpenCV, Tesseract OCR, PyTorch, and TensorFlow to extract ID-card data and detect faces behind FastAPI, Flask, and Django services, reducing manual onboarding time across bank branches.

Tech: Python, OpenCV, Tesseract OCR, PyTorch, TensorFlow, FastAPI, Flask, Django, PostgreSQL, AWS

Junior Research Analyst — Economic Research Group

Apr 2020 – Aug 2020

- Analyzed mobile-app data for Romoni, an on-demand beauty-services marketplace, and built regression-based sales forecasts that informed pricing and marketing decisions (Python, Pandas, scikit-learn).

Founder & CEO — Shikhte Chai (education initiative)

Apr 2016 – Jan 2020

- Founded and ran an education initiative in Bangladesh that taught students technology and entrepreneurship, leading strategy, operations, and program design.

PUBLICATION & PROJECT

- **Publication:** [Predicting the Result of a Cricket Match by Applying Data Mining Techniques](#) (ResearchGate, 2020): predicted one-day match winners using Recursive Feature Elimination with Decision Tree, Random Forest, and XGBoost models.
- **KO, a Bangla programming language:** Designed and built a programming language with Bangla-language syntax in Python, so native speakers and children can learn to code in their first language.

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science and Engineering — North South University, Bangladesh

2015 – 2019

Relevant coursework: Artificial Intelligence, Pattern Recognition & Neural Networks, Probability & Statistics, Algorithms

- Neo4j Certified Professional, Neo4j, 2025
- Data Science, Stanford University (Prof. Jennifer Widom), 2024: regression, classification, clustering